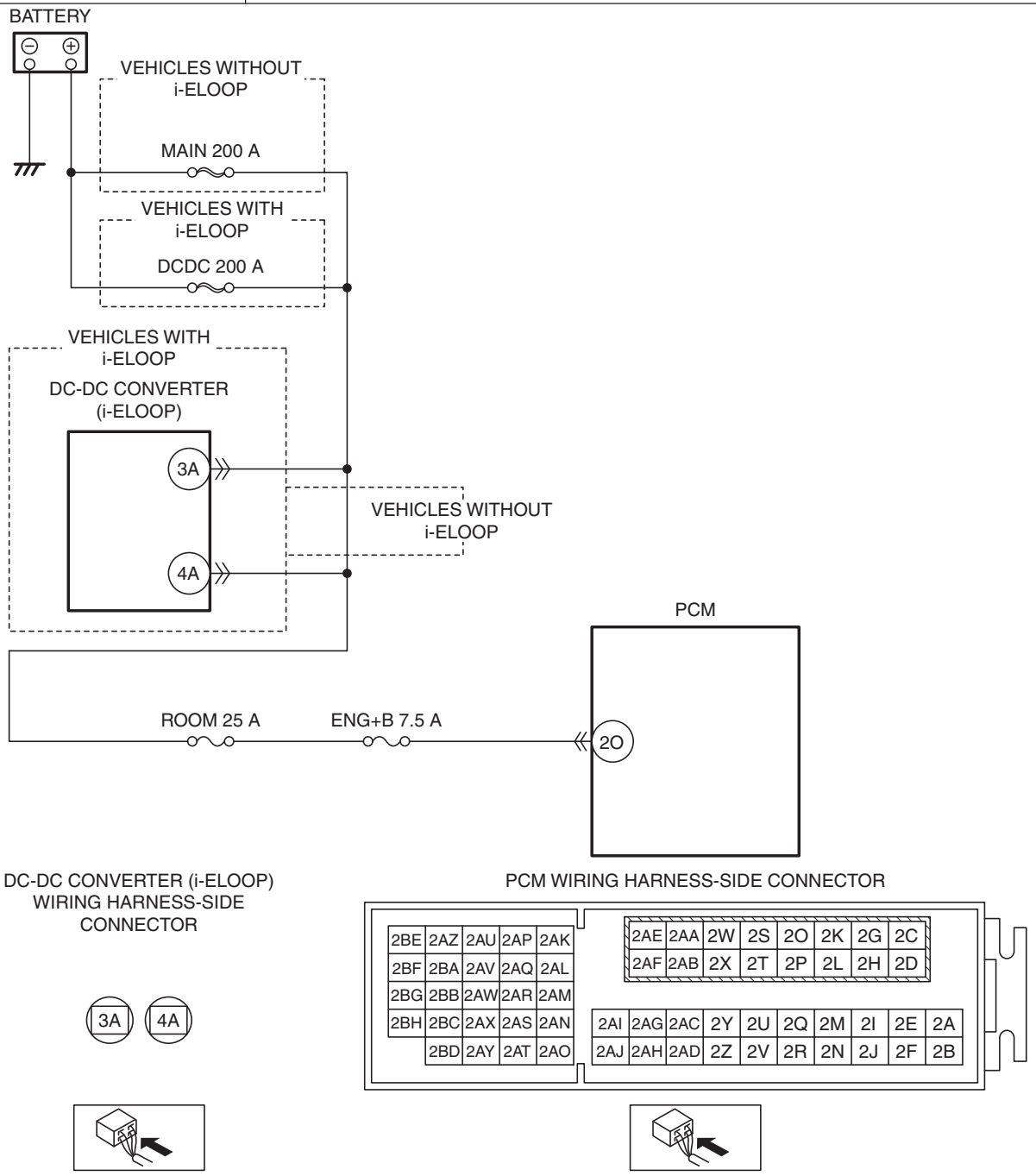
DTC U0401:00 [DSC HU/CM]

id040262887900

DTC	U0401:00	Signal error from PCM
DETECTION CONDITION		• Error detected in signal from PCM via CAN communication
FAIL-SAFE FUNCTION		Refer to DTC TABLE. (See DTC TABLE [DSC HU/CM].)
POSSIBLE CAUSE		• DTCs are stored in the PCM • DTCs are stored in the TCM — Short to ground in wiring harness between MAIN 200 A fuse and PCM terminal 2O — MAIN 200 A fuse and/or ROOM 25 A fuse and/or ENG+B 7.5 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and PCM terminal 2O • Short to ground or open circuit in backup voltage circuit (with i-ELOOP) — Short to ground in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and PCM terminal 2O — ROOM 25 A fuse and/or ENG+B 7.5 A fuse malfunction — Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and PCM terminal 2O • PCM malfunction • TCM malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY DTCs** • Using the M-MDS, perform the PCM and TCM DTC inspection. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (CW6A-EL)].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].) • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (CW6A-EL)].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (EW6A-EL)].) DTC troubleshooting completed, then go to the next step.
		No	**With i-ELOOP:** • Go to Step 3. **Without i-ELOOP:** • Go to the next step.
2	**INSPECT BACKUP VOLTAGE CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Reconnect all disconnected connectors. • Access the VPWR PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Verify the VPWR PID value. • Is the VPWR PID value **B+**?	Yes	Go to Step 4.
		No	Inspect the MAIN 200 A fuse, ROOM 25 A fuse and ENG +B 7.5 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and PCM terminal 2O. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the malfunctioning fuse. • If the fuse is damaged: — Replace the malfunctioning fuse. • If all fuses are normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and PCM terminal 2O. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 4.

Step	Inspection		Action
3	**INSPECT BACKUP VOLTAGE CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Reconnect all disconnected connectors. • Access the VPWR PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Verify the VPWR PID value. • Is the VPWR PID value **B+**?	Yes	Go to the next step.
		No	Remove the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Inspect the ROOM 25 A fuse and ENG+B 7.5 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between ROOM 25 A fuse and PCM terminal 2O. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. • Replace the malfunctioning fuse. • If the fuse is damaged: — Replace the malfunctioning fuse. • If all fuses are normal: — Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and PCM terminal 2O. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Go to the next step.
4	**DETERMINE IF MALFUNCTION IS PCM, TCM** • Inspect the PCM. (See PCM INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See PCM INSPECTION [SKYACTIV-D 1.5]) • Is the PCM normal?	Yes	Go to the next step.
		No	Replace the PCM, then go to the next step. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5	**VERIFY THAT THE SAME DTC IS NOT PRESENT** • Clear the DTCs from the memory. (See CLEARING DTC [DSC HU/CM].) • Are the same DTCs present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the control valve body, then go to the next step (See CONTROL VALVE BODY REMOVAL/INSTALLATION [CW6A-EL].) (See CONTROL VALVE BODY REMOVAL/INSTALLATION [EW6A-EL].) Go to the next step.
		No	Go to the next step.
6	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [DSC HU/CM].)
		No	DTC troubleshooting completed.